Selim Amrari Mariette

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Sr Data Scientist with 6 years of experience working at top-ranking startups. My key areas of specialization include Image processing, Multivariate time series forecasting, Text classification, Geospatial processing

Education

Master of Science — September 2015 - February 2018

Modeling and environmental monitoring
"École Polytechnique fédérale de Lausanne"

Core experience

Freelance - Sr Data Scientist — June 2025 - Now

- Product: built a decision-support dashboard for visualizing statistics and predicting customer flow for restaurants.
  Multivariate time series forecasting, Deep Learning, Bayesian methods

5-months career break to enjoy fatherhood - Apprentice Dad — January 2025 - May 2025

Calyps/Saniia (spin-off) - Sr Data Scientist — October 2021 - November 2024

- **Product**: Took leadership on hospital emergency patient flow forecasting product. Brought significant performance improvement through feature engineering, critical bugs identification and external data sources ingestion. Decreased overall prediction error by about 15%.
  Multivariate time series forecasting, Transformers Network/Deep Learning, Web Scraping
- **Customer relationship management:** conducted weekly meetings with product users
- Created three PoCs for customers demand forecasting and designed MVPs
  Transformers Network/Deep Learning
- Designed PoCs focused on NLP tasks such as automation of billing classification using text classification
  Transformers Network/Deep Learning, Hugging Face
- **Infrastructure**: Built a containerized infrastructure for all the company products, reducing deployment failures and improving scalability. Mentored the team on Docker.
  Docker
- Led technical interviews and designed assessments for data scientist positions

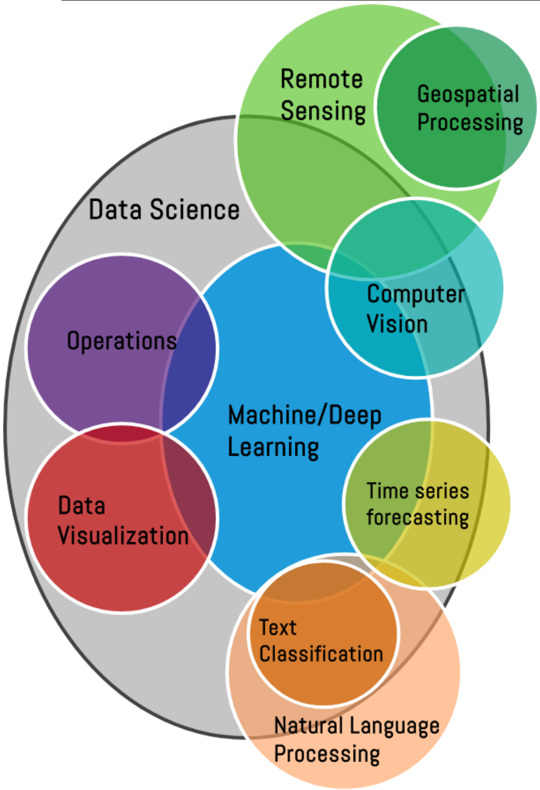
Gamaya - Sr Data Scientist — February 2019 - September 2021

- Product: **Developed a deep learning model, used by all the team, and combining recurrent and convolutional architecture in order to achieve spatio-temporal analysis of satellite images**
  Anomaly Detection, Sequential and Convolutional Neural Networks/Deep Learning
- Infrastructure/Product: Built an AI product dedicated to large scale sugarcane crop monitoring, estimating growth and detecting anomaly, used by customers to monitor thousands of parcels.
  Sequential and Convolutional Neural Networks/Deep Learning
- Infrastructure: Implemented a drone/satellite images alignment (coregistration) tool as a preprocessing building block.
  Computer vision algorithms
- Infrastructure: Engineered an automated cloud detection pipeline utilizing a deep learning model and processing 100% of incoming satellite imagery.
  Convolutional Neural Networks
- PoC: Conducted tests of Computer vision algorithms for sowing lines detection.
  Image processing approaches: Dynamic thresholding, Delineation, Clustering
- Mentored a Master-Intern on bio-physical parameters inversion project and supervised two products developed with subcontractors.

IFREMER - Research Engineer — August 2017 - September 2018

- Built a satellite-based bathymetry retrieval model delivering regular data for lagoon hydrodynamic modeling on thousands squared kilometers of coastal areas.
  Robust Regressions, Probabilistic mixtures models

Technical skills and programming tools



Broad expertise in M-L and Deep Learning:
TensorFlow, Keras, PyTorch Lightning, Pymc3, Scikit, Statsmodel

Natural Language Processing:
NLTK, Gensim, Huggingface

Image Processing and Computer Vision:
OpenCV, Scikit-image, photogrammetry

Remote Sensing and Geospatial Processing:
GDAL, OGR, Geoda, Geopandas, Qgis

Data handling and visualization:
Pandas, Dask, Xarray, Bokeh, streamlit, holoviews, bokeh

Expertise in ML Ops and Cloud:
Docker, Git, DVC, GCP, AWS, MLflow, Web Scraping, Linux

Additional Experience

Multivariate time series forecasting projects:

- ★ **Electricity demand forecasting**: presentation and demo
- ★ **Designed decision-support dashboard for restaurant customer flow forecasting**: presentation and demo
- ★ Built Auto-Updating COVID-19 Predictions Web Dashboard App: project page

Computer Vision and Remote sensing projects:

- ★ Developed Satellite Derived Bathymetry model optimized for lagoon ecosystem: presentation
- ★ Classified of urban tree species based on hyperspectral and LIDAR data : presentation and repo
- ★ Achieved Road segmentation from aerial imagery based on convolutional neural networks : report
- ★ **Designed a deep leaning model able to ingest raw satellite time series for change detection**: repo

Natural Language Processing project:

- ★ Automated papers assignation to journal using LLM and Text classification: presentation and repo
- ★ **Leveraged latent space from pretrained LLM to achieve zero-shot classification** : presentation and repo

Classical projects:
- ★ Customer churn detection

Language

French: Native language
English: Advanced
German: basic knowledge

Personal details

Happy young dad
Nationalities: Swiss, French
Birth: 08 november 1990 in Fribourg
Kitesurfer for 10 years, now surviving on wingfoil in Swiss Lakes
Definitely team ski – sorry, snowboarders
Ran a 40km trail once, hoping to catch up with my old shape… still walking, waiting for it